

Field development — concept screening, cost & economics

The oil-and-gas concept-select twin of the floating-wind LCOE work: screen an offshore prospect across four decision axes (schedule, rig-days, driver-level CAPEX and lifecycle intervention), rank host concepts (TLP / spar / semi / FPSO / subsea tieback), and run pre-tax discounted-cash-flow economics on the survivors. All parameters are generic Gulf-of-Mexico defaults from a reviewable YAML; every field is an INPUT — no reservoir, flow-assurance or structural simulation is solved. Rig demand is reported in days, never \$/day.

4

decision axes

10

GoM benchmark fields

~\$55/bbl

reference break-even

WHAT YOU GET

- Four-axis concept screen (schedule weeks, development rig-days, driver-level CAPEX, intervention index) over TLP / spar / semi / FPSO / subsea-tieback hosts
- CAPEX ranges anchored to 10 GoM reference fields, scaled by a 0.6-power capacity law and a depth factor, with distance-banded subsea-tieback logic
- Pre-tax DCF economics live: CAPEX selector and oil-price slider drive NPV across break-even, with hyperbolic / harmonic Arps decline and five fiscal regimes in the same engine
- Field-development layout schematic generator plus a structured catalog of worked reference developments
- Honesty contract: rule / parameter screening only — rate and well count are inputs, rig demand in days, never \$/day